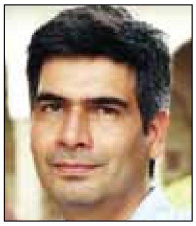
the CBDCs would flow through the economy, and what role banking and financial institutions, payment services, and application providers play in the ecosystem.

"Digital rupee will complement the existing payment modes and can make a significant impact in a few notable areas like offline payments, programmable or targeted payments (subsidies), cross border payments and remittances." *— **Mihir Gandhi, Partner and Leader - Payments Transformation, PricewaterhouseCoopers Pvt Ltd (PwC)***

"While we have digital payment systems today, including immediate settlements through IMPS, UPI, etc, they are not universal and sometimes carry high fees or transaction costs (particularly cross-border). CBDCs will allow cross-border payments at fractional costs and deliver unprecedented impact." *— **Rajesh Lalwani, CEO, Scenario Consulting Pvt Ltd***



Blockchain and other technologies

India's CBDC is likely to use DLT, most likely a private blockchain handled by the government. This would enable RBI to have good control over the blockchain. Sankar explained in his speech that a blockchain can be maintained without native currency, if transactions are authenticated centrally. It is possible to maintain accounts and reward work with legal tender currency too, making the DLT useful for several purposes without any private currency.

The iSPIRT Foundation earlier launched the Bharat Distributed Ledger (BADAL) for establishing trust for Indian B2B commerce. The BADAL protocol uses a private-public ledger (PPL) substrate, which might be useful for the RBI to launch CBDC.

Garg suggested in another media interview that it would be much easier, convenient, and cost-effective to dematerialise the physical currency on the lines of how equity, bonds, and other securities have been dematerialised, and use it to usher in digital currency in India.

According to Eswar Prasad, Professor at Cornell University, retail CBDC could take two forms—value based or token based. In a Bloomberg Quint interview, he said, "One could be a prepaid card, except it would be an account with a prepaid balance that you would have on your phone. A second form is where a CBDC essentially functions as a token in a payment system. This would take the form of a digital wallet where one could hold CBDCs, which would essentially be like accounts, but non-interest-bearing accounts."

"Blockchain is one of the many distributed ledger technologies available for launching a CBDC, however not the only technology available. We will need to evaluate available technologies on multiple criteria like accessibility, privacy, scalability, performance, resiliency, finality etc. Decision needs to be taken based on the model selected, if there should be a centralised technology or decentralised technology," concludes Gandhi.

Pitfalls to be avoided

While designing its CBDC framework, India must take into consideration some potential problems, and try to design the solutions into the system right in the beginning:

Potential bank run. A bulk shift from commercial to central banks might lead to a kind of financial crisis. To avoid this, central banks must limit the amount of money that can be held in CBDC accounts, and make sure digital wallets are maintained by commercial banks.

Strain on the central bank. Full control in the hands of central banks might lead to a strain on their network infrastructure and make the central bank vulnerable to security threats.

Private innovation. Reduced role of commercial banks might kill private innovation in the banking sector. To avoid this, countries like Sweden and China are trying out a two-tier CBDC, where the central bank provides the payment infrastructure in the backend. In the front-end, commercial banks innovate and provide effective payment mechanisms to users.

Policy making challenges. CBDC, with the direct nature of its transactions, will have a dynamic impact on the economy, and macroeconomic policymaking will be more challenging than ever before!

Energy consumption. Blockchain based cryptographic networks require immense computing power, and many of them, including the Bitcoin network, have come under heavy criticism for their socially wasteful energy usage. However, high energy consumption is not intrinsic to all blockchain network architectures. The amount of energy consumed by a blockchain network depends on its consensus mechanism, that is, what information is added to the network ledger. India must optimise this, such that it does not have a deep environmental impact.

India is in for a very interesting ride, which involves not only designing and developing the infrastructure, and rolling out the digital currency in a phased manner, but also clearing misconceptions and creating awareness about digital currencies—what is accepted and what is not. We might see wholesale CBDC in action first, followed by retail CBDC in a simple form, with more features being added as we go along. EFY